AVANAGANTI BHANU PRASAD

GENERATIVE AI & AI AUTOMATION ENGINEER · RAG · LLM AGENTS · PYTHON BACKENDS

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SUMMARY

Final-year B.Tech (CSE — AI & Machine Learning) engineer building LLM systems that are measured and controlled, not just demoed: RAG pipelines with hybrid retrieval and citation-level grounding, and multi-agent platforms with schema-validated outputs and fallback.

EDUCATION

Aurora University, Hyderabad — B.Tech, Computer Science & Engineering *(AI & Machine Learning)* **2023–2027 (Final Year)**

CGPA 4/5 · Intermediate (Class 12): 922/1000 · SSC (Class 10): CGPA 10/10

TECHNICAL SKILLS

Languages & Core: Python, TypeScript/JavaScript, SQL, Workflow Automation (n8n), Data Structures & Algorithms, OOP, Git/GitHub

Generative AI & LLM: RAG pipelines, prompt engineering, embeddings, hybrid retrieval (FAISS + BM25), rank fusion, groundedness & citation validation, structured JSON output, abstention guardrails, multi-provider LLM routing (Groq, Gemini, OpenAI-compatible APIs)

Backend & APIs: FastAPI, Uvicorn, REST API design, Pydantic validation, retries & timeout control, third-party API integration (SerpApi, Sarvam AI, Groq)

DevOps & Cloud: Docker, GitHub Actions CI, pytest, Google Vertex AI, Google Cloud Storage, Render / Hugging Face Spaces

PROJECTS

Voice-Enabled RAG over Indic Text — Aug 2026

Python · FAISS · BM25 · Groq · Sarvam STT · Pydantic · Docker **github.com/bhanu964/voice-rag-indic**

- Built an end-to-end Hindi/English voice-to-answer RAG pipeline — speech-to-text, embedding, retrieval, generation and a grounded, cited answer — over a 20K-passage corpus.
- Engineered hybrid retrieval pairing a FAISS index with a custom BM25 index fused by Reciprocal Rank Fusion, holding retrieval latency at 23 ms P50.
- Implemented guardrails that abstain rather than guess, including post-generation citation validation that rejects any claim not traceable to a retrieved chunk.

CareerVolt — Multi-Agent LLM Platform — Aug 2026

Winner — Algorand x402 Web3 Hackathon

TypeScript · Node.js · React · Gemini / Groq · Algorand **github.com/bhanu964/careervolt**

- Designed five task-specialised LLM agents — resume analysis, JD matching, career roadmap, job-market mapping and a copilot — each exposed as an independent HTTP endpoint.
- Built a provider-agnostic LLM bridge routing to Gemini, Groq or any OpenAI-compatible endpoint, enforcing JSON-only responses against a declared schema, with a rule-engine fallback that keeps every agent functional without an LLM key.
- Automated document intake by extracting text from PDF and DOCX resumes, then driving LLM-based structured extraction of skills, roles and education.

Face-Evidence Verification Pipeline — Sep 2026

Python · FastAPI · ONNX Runtime · Google Lens API · Docker · GitHub Actions **github.com/bhanu964/hh-goa-task3**

- Engineered a face-verification pipeline combining ONNX Runtime embeddings with live reverse-image search through the Google Lens API, served behind a FastAPI upload interface.
- Produced a deterministic audit record for every decision — source, URL, image hash, similarity score and threshold — storing the hash only, so biometric data never leaves the host.
- Shipped as a modular Python package with Docker and a GitHub Actions CI matrix running pytest across Python 3.11–3.13.

Genetic Disorder Prediction — Cloud-Trained ML Pipeline — Apr 2026

Python · scikit-learn · Google Vertex AI · Google Cloud Storage **github.com/bhanu964/-genetic-disorder-ml**

Compared Random Forest, SVM, Logistic Regression and KNN under K-fold cross-validation, training on Google Vertex AI and persisting the selected model to Google Cloud Storage.

ACHIEVEMENTS, RESEARCH & CERTIFICATIONS

- **Winner — Algorand x402 Web3 Hackathon**
- **Research presentation (oral)** — "AI-Powered Credit Card Fraud Detection," ICASET 2026, Vidya Vihar Institute of Technology, Purnea.
- **AI–Machine Learning Engineer** (150 hours) — Reliance Foundation Skilling Academy / Skill India Digital Hub, 2026